

CUSTOMER CHURN & RETENTION ANALYTICS

A Data Analytics, SQL, Machine Learning and Streamlit Project
IBM SkillBuild Data Analytics With AI Internship

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Technologies Used: Python, SQL, Machine Learning, Streamlit

ABSTRACT

Customer retention is a significant concern for telecom companies, as the loss of existing customers can influence business performance and customer relationships. This project examines telecom customer data to understand the characteristics and behaviour associated with churn. It also aims to divide customers into meaningful segments based on their estimated likelihood of churn and generate insights that can assist in planning retention activities.

The analysis is based on a Telecom Churn Dataset containing information related to customer demographics, location, registration, salary, communication activity, data consumption, and churn status. Python was used for data preparation, exploratory analysis, visualization, KPI development, and machine learning. SQL was used to perform structured analysis of customer and churn information, and Streamlit was used to create an interactive dashboard for presenting the findings.

Machine learning techniques were applied to calculate an estimated churn probability for individual customers. These probabilities were then used to group customers into relative risk categories. The project follows a structured **Data** → **Information** → **Insights** → **Decision** → **Action** approach, connecting technical analysis with potential business applications.

Overall, the project demonstrates how customer data can be transformed into useful analytical information for studying churn behaviour, identifying customer groups that may require further attention, and supporting informed customer retention planning.

INTRODUCTION

Customer retention is an important aspect of the telecom industry because companies need to maintain strong relationships with existing customers while continuing to attract new ones. When customers discontinue a telecom service, the organization may lose recurring revenue and valuable customer relationships. Understanding customer behavior can therefore help businesses identify patterns that are associated with customer churn.

This project focuses on analyzing telecom customer data to gain a better understanding of churn behavior. The dataset contains information about customer demographics, location, registration details, estimated salary, communication activity, data usage, and churn status. By examining these variables, the project aims to identify differences between customers who remain with the service and those who have churned.

The project combines several analytical techniques to provide a complete view of the problem. **Python** is used for data cleaning, exploratory data analysis, visualization, KPI calculation, and machine learning. **SQL** is used to organize and query customer information for business analysis, while **Streamlit** is used to present the findings through an interactive dashboard.

Machine learning is also incorporated to estimate the probability of customer churn and divide customers into relative predicted-risk groups. These results can help identify customer segments that may require further investigation or retention-focused attention.

The overall project follows the **Data → Information → Insights → Decision → Action** framework. This approach connects technical data analysis with practical business understanding and demonstrates how customer data can be transformed into useful information for supporting customer retention planning.

PROBLEM STATEMENT

Telecom companies handle a large number of customers with different demographic characteristics, locations, and service usage patterns. Some customers may discontinue their services, resulting in customer churn. However, identifying the customer characteristics and usage patterns associated with churn can be challenging when large amounts of customer data are involved.

The problem addressed in this project is to **“analyse telecom customer data to understand the factors and patterns associated with customer churn, identify customer groups with different levels of predicted churn risk, and generate useful insights that can support customer retention planning”**.

The project uses data related to customer demographics, telecom partners, location, registration, estimated salary, calls, SMS activity, data usage, and churn status. Through data analysis, SQL queries, machine learning, and an interactive dashboard, the project aims to transform this information into meaningful insights for business analysis.

The central question of the project is:

“How can telecom customer data be analyzed to understand churn patterns, estimate relative customer churn risk, and support data-driven customer retention strategies?”

OBJECTIVES

The main objective of this project is to analyze telecom customer data and develop a data-driven approach for understanding customer churn and supporting retention planning.

The specific objectives are:

1. **To examine customer churn patterns** across demographic, geographic, and service-related characteristics.
2. **To identify differences between churned and retained customers** based on variables such as age, dependents, salary, calls made, SMS sent, and data usage.
3. **To calculate key business performance indicators (KPIs)** such as total customers, churned customers, churn rate, and retention rate.
4. **To analyze churn using SQL** by examining customer groups, telecom partners, gender, states, usage patterns, and predicted-risk segments.
5. **To develop machine learning models** that estimate the probability of customer churn using available customer information.
6. **To classify customers into relative predicted-risk segments** based on their estimated churn probabilities.
7. **To develop an interactive Streamlit dashboard** that presents important KPIs, churn patterns, risk segments, and customer-level information.
8. **To generate actionable business insights** that can help organizations understand customer churn and plan appropriate retention strategies.
9. **To demonstrate the complete analytics workflow** from **Data** → **Information** → **Insights** → **Decision** → **Action** using Python, SQL, Machine Learning, and data visualization.

DATASET DESCRIPTION

The dataset used for this project is the **Telecom Churn Dataset**, obtained from Kaggle. It contains customer-level information that can be used to study customer characteristics, service usage patterns, and churn behaviour.

The dataset used in this project contains **243,553 customer records and 14 variables**. Each record represents an individual telecom customer.

Dataset Source:

<https://www.kaggle.com/datasets/suraj520/telecom-churn-dataset>

Dataset Variables

Variable	Description
customer_id	Unique identification number assigned to each customer
telecom_partner	Telecom partner associated with the customer
gender	Gender of the customer
age	Age of the customer
state	State in which the customer is located
city	City in which the customer is located
pincode	Postal code associated with the customer's location
date_of_registration	Date on which the customer registered for the service
num_dependents	Number of dependents associated with the customer
estimated_salary	Estimated salary of the customer
calls_made	Number of calls made by the customer
sms_sent	Number of SMS messages sent by the customer
data_used	Amount of data used by the customer
churn	Indicates whether the customer has churned

Target Variable

The primary target variable for the machine learning analysis is:

churn

It is a binary variable:

0 = Customer retained

1 = Customer churned

Data Preparation

Before analysis, the dataset was inspected for missing values, duplicate records, incorrect data types, and invalid values. The registration-date column was converted into a suitable date format. Duplicate customer IDs were checked, and invalid negative values in the data_used variable were treated as data-quality issues and replaced using the median of valid data-usage values.

The cleaned dataset was then used for **exploratory data analysis, KPI analysis, SQL analysis, machine learning, risk segmentation, and dashboard development.**

TECHNOLOGIES

The following technologies and tools were used to develop the **Customer Churn & Retention Analytics** project:

1. Python

Python was used as the primary programming language for data processing, analysis, visualization, and machine learning.

2. Pandas

Pandas was used to load, clean, transform, and analyze the customer dataset. It was also used for grouping customers and calculating churn-related KPIs.

3. NumPy

NumPy was used for numerical calculations and handling numerical data during the data-cleaning and machine-learning processes.

4. Matplotlib

Matplotlib was used to create charts and visualizations for understanding customer churn patterns and presenting analytical results.

5. Seaborn

Seaborn was used for statistical visualizations, including comparative charts and correlation analysis.

6. Scikit-learn

Scikit-learn was used for machine learning. It supported data preprocessing, feature encoding and scaling, model training, and evaluation. **Logistic Regression** and **Random Forest** were implemented for churn prediction.

7. SQLite

SQLite was used as the database system for storing customer information and performing SQL-based analysis. The project generates a database named: `customer_churn.db`

8. SQL

SQL queries were used to calculate and analyse customer KPIs, churn by telecom partner, gender, state, customer usage, and predicted-risk segments.

9. Streamlit

Streamlit was used to develop the interactive dashboard. It allows users to view KPIs, charts, customer segments, usage information, and predicted-risk information through a web-based interface.

10. Visual Studio Code

Visual Studio Code was used as the primary development environment for writing and executing the Python programs, managing project files, and running the Streamlit dashboard.

Technology Stack Summary

Python

|

└ Pandas / NumPy → Data Processing

└ Matplotlib / Seaborn → Visualization

└ Scikit-learn → Machine Learning

↓

SQLite / SQL

↓

Streamlit

↓

Interactive Dashboard

METHODOLOGY

The methodology used in this project follows an end-to-end analytical workflow that converts raw telecom customer data into meaningful information, insights, and business-oriented actions. The project follows the **Data** → **Information** → **Insights** → **Decision** → **Action** framework.

7.1 OVERALL WORKFLOW

The complete workflow of the project is:

Raw Dataset



Data Inspection



Data Cleaning and Preparation



Exploratory Data Analysis



KPI and Churn Analysis



Churn Driver Analysis



Machine Learning



Relative Predicted-Risk Segmentation



SQL Analysis



Streamlit Dashboard



Business Insights and Retention Actions

7.2 DATA COLLECTION AND INSPECTION

The Telecom Churn Dataset was obtained from Kaggle and loaded into Python using Pandas. The dataset was initially examined to understand its structure, number of records, variables, data types, missing values, duplicate records, and possible invalid values.

The target variable, **churn**, was treated as a binary classification variable where 0 represents a retained customer and 1 represents a churned customer.

7.3 DATA CLEANING AND PREPARATION

The dataset was cleaned before performing analysis. The main preparation steps included:

Checking for missing values and duplicate records.

Checking for duplicate customer IDs.

Converting date_of_registration into a suitable datetime format.

Checking the churn variable for valid binary values.

Identifying invalid negative values in data_used.

Replacing invalid negative data-usage values with the median of valid values.

Creating additional analytical groups for age, calls made, and data usage.

These steps helped prepare the dataset for reliable analysis and machine learning.

7.4 EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand customer characteristics and identify patterns associated with churn. The analysis examined churn across variables such as:

Telecom partner

Gender

Age groups

Number of dependents

State

Calls made

SMS sent

Data used

Numerical characteristics were also compared between churned and retained customers. Visualizations using Matplotlib and Seaborn were used to make patterns and differences easier to interpret.

The EDA is descriptive in nature, so observed relationships are treated as associations rather than proof of causation.

7.5 KPI AND CHURN ANALYSIS

Key performance indicators were calculated to convert customer-level data into business-oriented information. These included:

Total customers

Churned customers

Churn rate

Retention rate

Churn by telecom partner

Churn by gender

Churn by state

Usage comparisons between churned and retained customers

The results were also exported into CSV files for further analysis and reporting.

7.6 MACHINE LEARNING

Machine learning was used to estimate the probability of customer churn. Customer demographic, location, and usage-related variables were used as model inputs.

Two classification algorithms were implemented:

Logistic Regression

Random Forest

The dataset was divided into training and testing sets using an **80:20 split**, with stratification based on the churn variable. Numerical variables were processed using median imputation and scaling, while categorical variables were handled using imputation and one-hot encoding.

The models were evaluated using classification metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix.

7.7 CUSTOMER RISK SEGMENTATION

The estimated churn probabilities generated by the machine learning model were used to create **relative predicted-risk segments**.

Customers were divided into three groups:

Lower Predicted Risk

Medium Predicted Risk

Higher Predicted Risk

The segmentation was created using relative probability distribution within the analyzed customer population. Therefore, these categories represent **relative predicted risk** rather than guaranteed future churn.

7.8 SQL ANALYSIS

The cleaned and processed customer data was stored in a SQLite database named `customer_churn.db`. SQL queries were then used to perform structured business analysis.

The SQL analysis covered:

Overall customer and churn KPIs

Churn by telecom partner

Churn by gender

State-level churn analysis

Usage comparison by churn status

Predicted-risk segment analysis

This step demonstrated how SQL can be used to transform customer data into structured business information.

7.9 STREAMLIT DASHBOARD

A Streamlit application was developed to present the major analytical results through an interactive dashboard. The dashboard includes KPI cards, filters, churn analysis charts, risk-segment information, usage comparisons, and customer-level information for customers in the higher predicted-risk segment.

The dashboard provides a user-friendly interface for exploring the results without directly working with the underlying Python code or SQL queries.

7.10 BUSINESS INSIGHT AND ACTION

The final stage of the methodology connects the analytical results with potential business applications. The identified churn patterns and relative predicted-risk segments can be used to support further investigation and retention planning. Thus, the methodology connects:

Data → Information → Insights → Decision → Action

This provides a structured approach for transforming raw telecom customer records into useful information that can support customer retention analysis.

DATA CLEANING

The dataset was cleaned and prepared before performing exploratory analysis, SQL analysis, and machine learning. The purpose of this stage was to identify data-quality issues and convert the raw dataset into a suitable format for further analysis.

8.1 DATA INSPECTION

The dataset was first inspected to understand its structure, variables, data types, missing values, duplicate records, and invalid values. The dataset contained **243,553 customer records and 14 columns**.

The initial inspection showed that there were **no missing values and no duplicate rows**.

8.2 DATE CONVERSION

The `date_of_registration` column was converted into a proper datetime format. This ensured that the registration information could be handled correctly during analysis.

```
df["date_of_registration"] = pd.to_datetime(  
    df["date_of_registration"],  
    errors="coerce"  
)
```

8.3 DUPLICATE CUSTOMER CHECK

Although no duplicate rows were found during the initial inspection, duplicate customer IDs were also checked to ensure that each customer was represented only once.

```
df = df.drop_duplicates(subset="customer_id")
```

8.4 HANDLING INVALID DATA USAGE VALUES

During the data-quality inspection, negative values were identified in the `data_used` column. Since negative data usage is not considered a valid usage value, these records were treated as invalid.

The invalid values were first converted to missing values and then replaced using the median of the valid data-usage values.

```
df.loc[df["data_used"] < 0, "data_used"] = np.nan  
df["data_used"] = df["data_used"].fillna(  
    df["data_used"].median()  
)
```

This approach allowed the affected records to remain in the dataset without using an invalid negative value.

8.5 VALIDATION OF CHURN VALUES

The churn column was used as the target variable for the analysis. Only valid binary values, 0 and 1, were retained.

```
df = df[df["churn"].isin([0, 1])]
```

Here:

0 represents a retained customer.

1 represents a churned customer.

8.6 CREATION OF ANALYTICAL GROUPS

Additional groups were created to support segment-level analysis. Customers were grouped into age categories, while calls made and data usage were divided into quartile-based groups.

These derived groups helped compare customer behavior across different ranges and made the exploratory analysis easier to interpret.

8.7 FINAL PREPARED DATASET

After completing the cleaning and preparation process, the dataset was used for:

- Exploratory Data Analysis
- KPI and churn analysis
- Churn driver analysis
- Machine learning
- Relative predicted-risk segmentation
- SQL analysis
- Streamlit dashboard development

The cleaning process ensured that the subsequent analysis was performed using a consistent and structured dataset.

EDA(EXPLORATORY DATA ANALYSIS)

Exploratory Data Analysis (EDA) was performed to understand the distribution of customer characteristics and identify patterns associated with customer churn. The analysis focused on demographic, geographic, telecom-partner, and service-usage variables.

9.1 CHURN DISTRIBUTION

The churn variable was examined to understand the overall distribution of retained and churned customers. This provided an initial view of the customer base and established the basis for further churn analysis.

9.2 CHURN BY TELECOM PARTNER

Customer churn was analyzed across different telecom partners. This comparison helps identify differences in observed churn rates between partner groups and provides a basis for further investigation of partner-level customer behavior.

9.3 CHURN BY GENDER

The dataset was grouped by gender to compare the observed churn patterns between customer groups. This analysis helps understand whether the distribution of churn differs across the available gender categories.

9.4 CHURN BY AGE GROUP

Customers were divided into age groups to examine churn patterns across different age ranges. The following groups were created:

- Under 18
- 18–25
- 26–35
- 36–45
- 46–55
- 56–65
- 65+

The churn distribution across these groups was then compared using visualizations.

9.5 CHURN BY NUMBER OF DEPENDENTS

The number of dependents was analyzed in relation to customer churn. This helped examine whether customers with different dependent counts showed different observed churn patterns.

9.6 CHURN BY STATE

State-level analysis was performed to examine geographic differences in customer churn. Churn rates were calculated for different states, allowing customer groups from different locations to be compared.

9.7 USAGE-BASED ANALYSIS

Customer activity was examined using:

- calls_made
- sms_sent
- data_used

Customers were also grouped into usage ranges using quartile-based segmentation. These groups were compared to understand how communication and data-usage patterns differed across customers.

9.8 COMPARISON OF CHURNED AND RETAINED CUSTOMERS

Numerical characteristics were compared between customers who had churned and customers who had remained with the service. The analysis included variables such as:

- Age
- Number of dependents
- Estimated salary
- Calls made
- SMS sent
- Data used

This comparison helped identify observable differences between the two churn-status groups.

9.9 CORRELATION ANALYSIS

Correlation analysis was also performed on numerical variables to examine the relationships between customer characteristics and usage-related variables. A correlation heatmap was used to make these relationships easier to interpret.

9.10 EDA INTERPRETATION

The EDA stage converts the raw customer dataset into descriptive information that can be used for further analysis. The observed differences across customer groups were used as inputs for KPI analysis, churn-driver analysis, machine learning, and customer risk segmentation.

It is important to note that the EDA findings describe **associations and patterns in the available data**. They should not automatically be interpreted as evidence that a particular variable directly causes customer churn.

KPI ANALYSIS

Key Performance Indicator (KPI) analysis was performed to convert the customer-level dataset into meaningful business information. These KPIs provide a structured view of customer retention and churn and allow comparisons across different customer segments.

10.1 OVERALL CUSTOMER KPIs

The main KPIs calculated in the project were:

- **Total Customers** – Total number of customers included in the cleaned dataset.
- **Churned Customers** – Number of customers whose churn value is 1.
- **Churn Rate** – Percentage of customers who have churned.
- **Retention Rate** – Percentage of customers who have remained with the service.

The churn rate was calculated as:

$$\text{Churn Rate} = (\text{Churned Customers} / \text{Total Customers}) \times 100$$

The retention rate was calculated as:

$$\text{Retention Rate} = (\text{Retained Customers} / \text{Total Customers}) \times 100$$

These measures provide a high-level view of the customer base and its churn status.

10.2 CHURN BY TELECOM PARTNER

The dataset was grouped by telecom_partner to calculate the number of customers and observed churn rate for each telecom partner.

This analysis helps identify differences in churn patterns between partner groups and provides information that can be investigated further.

10.3 CHURN BY GENDER

Customer records were grouped by gender to calculate customer counts and observed churn rates for each available gender category.

This provides a segment-level view of the distribution of churn within the dataset.

10.4 CHURN BY STATE

State-level KPIs were calculated to examine geographic differences in customer churn. The analysis included customer count, churned customer count, and observed churn rate.

States with very small customer populations were treated carefully when interpreting the results, since percentages based on small groups may be less stable.

10.5 USAGE COMPARISON

Customer usage variables were also analyzed according to churn status. The main variables included:

- Calls made

- SMS sent
- Data used

Average usage values were compared between churned and retained customers to identify observable differences in customer activity.

10.6 KPI OUTPUT FILES

The calculated KPI results were saved as separate CSV files for further analysis and reporting:

- partner_kpi.csv
- state_kpi.csv
- age_kpi.csv

Additional SQL-based KPI outputs were also generated for structured analysis.

10.7 BUSINESS PURPOSE OF KPI ANALYSIS

The KPI analysis transforms the raw customer dataset into measurable business information. These indicators can be used to monitor churn, compare customer segments, identify areas requiring further investigation, and provide supporting information for customer retention planning.

MACHINE LEARNING

Machine learning was incorporated into the project to estimate the probability that an individual customer may churn. The purpose of this stage was to extend the descriptive analysis and provide a basis for creating relative predicted-risk customer segments.

1 1.1 TARGET VARIABLE

The target variable used for machine learning was **churn**:

- 0 – Customer retained
- 1 – Customer churned

1 1.2 FEATURES USED

The following customer attributes were used as input features:

- telecom_partner
- gender
- age
- state
- num_dependents
- estimated_salary
- calls_made
- sms_sent
- data_used

These variables represent customer demographic, geographic, and usage-related information.

1 1.3 DATA PREPROCESSING

Before model training, the features were divided into numerical and categorical variables.

Numerical features:

- Age
- Number of dependents
- Estimated salary
- Calls made
- SMS sent
- Data used

Numerical variables were processed using **median imputation and standard scaling**.

Categorical features:

- Telecom partner
- Gender
- State

Categorical variables were processed using **most-frequent-value imputation and one-hot encoding**.

This preprocessing was implemented as part of the machine learning pipeline.

1 1.4 TRAIN-TEST SPLIT

The dataset was divided into training and testing sets using an **80:20 split**. Stratification was applied using the churn variable so that the target classes were represented appropriately in both datasets.

1 1.5 MODELS USED

Two classification algorithms were implemented:

1. Logistic Regression

Logistic Regression was used as a classification model for estimating the probability of customer churn.

2. Random Forest

Random Forest was implemented as a second classification approach using multiple decision trees to model relationships between customer characteristics and churn.

1 1.6 MODEL EVALUATION

The models were evaluated using the following classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

A model comparison output was also generated to organize the evaluation results.

The actual metric values are stored in the generated **model_comparison.csv** file and should be used if numerical model results are required in the report.

1 1.7 CHURN PROBABILITY

The selected machine learning model was used to generate an estimated **churn probability** for each customer. This probability represents the model's estimated likelihood of churn based on the available customer information.

These probabilities were subsequently used in the customer risk-segmentation stage.

1 1.8 PURPOSE OF THE MACHINE LEARNING STAGE

The machine learning component provides an additional analytical layer to the project. Instead of only examining historical churn patterns, the project uses model-generated probabilities to organize customers into relative predicted-risk groups.

It is important to note that the predicted probabilities are **model estimates and are not guarantees that a particular customer will churn.**

RISK SEGMENTATION

Customer risk segmentation was performed after the machine learning stage to organize customers according to their estimated probability of churn. The purpose of this step was to make the model output easier to interpret and use for customer-level analysis.

12.1 CHURN PROBABILITY

The selected machine learning model generated a **churn probability** for each customer. This probability was stored in the `churn_probability` column and represents the model's estimated likelihood of churn.

12.2 RISK SEGMENTATION METHOD

Instead of using fixed probability thresholds, the project used **quantile-based segmentation** to divide customers into three groups based on their predicted churn probabilities.

The three segments are:

1. **Lower Predicted Risk**
2. **Medium Predicted Risk**
3. **Higher Predicted Risk**

The segmentation was created using the following approach:

```
df["risk_level"] = pd.qcut(
    df["churn_probability"],
    q=3,
    labels=[
        "Lower Predicted Risk",
        "Medium Predicted Risk",
        "Higher Predicted Risk"
    ],
    duplicates="drop"
)
```

This method divides the analyzed customer population into approximately equal-sized groups according to their predicted churn probabilities.

12.3 SEGMENT DISTRIBUTION

The resulting segmentation contained:

Risk Segment	Number of Customers
Lower Predicted Risk	81,185
Medium Predicted Risk	81,184
Higher Predicted Risk	81,184
Total	243,553

The three groups therefore contain approximately one-third of the analyzed customer population each.

12.4 INTERPRETATION

The **Higher Predicted Risk** segment contains customers with relatively higher model-generated churn probabilities within the analyzed population. This group can be used for further investigation and potential retention-focused analysis.

The **Medium Predicted Risk** segment represents customers whose estimated churn probabilities fall in the middle portion of the analyzed population.

The **Lower Predicted Risk** segment contains customers with relatively lower estimated churn probabilities.

These categories are **relative to the analyzed customer population**. They should not be interpreted as a guarantee that a customer will or will not churn.

12.5 BUSINESS USE

The risk segmentation provides a structured way to identify customer groups that may require different levels of attention. The higher predicted-risk segment can be examined further using customer characteristics, usage behavior, location, and other available information.

The resulting risk scores were saved in **customer_churn_risk_scores.csv** and were also used in the SQL analysis and Streamlit dashboard.

SQL ANALYSIS

SQL analysis was performed to organize the cleaned customer data into a structured database and generate business-oriented summaries. **SQLite** was used as the database system, and the database created for the project was named **customer_churn.db**.

13.1 DATABASE PREPARATION

The processed customer dataset was loaded into the SQLite database. SQL queries were then executed to analyze customer churn from different perspectives.

The SQL analysis focused on:

- Overall customer KPIs
- Churn by telecom partner
- Churn by gender
- State-level churn
- Usage comparison
- Predicted-risk segments

13.2 OVERALL KPI ANALYSIS

The first SQL query was used to calculate overall customer metrics such as:

- Total number of customers
- Number of churned customers
- Number of retained customers
- Churn rate
- Retention rate

This provided a summary view of the overall customer base.

13.3 CHURN BY TELECOM PARTNER

SQL GROUP BY operations were used to calculate customer counts and observed churn rates for each telecom partner.

This analysis provides a structured way to compare churn patterns across different partner groups.

13.4 CHURN BY GENDER

Customer records were grouped according to gender to calculate customer counts and observed churn information for each category.

This allowed the gender-based churn patterns identified during EDA to also be examined using SQL.

13.5 STATE-LEVEL CHURN ANALYSIS

SQL was used to calculate customer counts, churned customers, and observed churn rates by state.

A minimum customer-count condition was applied when identifying the top states by observed churn rate so that very small groups did not dominate the comparison.

13.6 USAGE ANALYSIS

SQL queries were also used to compare customer usage characteristics according to churn status. The analysis included:

- Calls made
- SMS sent
- Data used

This provided a structured summary of usage behaviour for churned and retained customers.

13.7 PREDICTED-RISK SEGMENT ANALYSIS

The customer risk-segmentation results generated by machine learning were also stored in the database. SQL was used to summarize customers across the three relative predicted-risk categories:

- Lower Predicted Risk
- Medium Predicted Risk
- Higher Predicted Risk

This connected the machine learning results with the database analysis.

13.8 SQL OUTPUT FILES

The results of the SQL queries were exported into separate CSV files:

- sql_overall_kpis.csv
- sql_partner_churn.csv
- sql_gender_churn.csv
- sql_state_churn.csv
- sql_usage_churn.csv
- sql_risk_segments.csv

These outputs were also used as supporting inputs for reporting and dashboard development.

13.9 PURPOSE OF SQL ANALYSIS

The SQL stage demonstrates how customer data can be stored, queried, grouped, and summarized using a relational database. It provides a structured approach for converting customer records into business-oriented information and complements the Python-based analysis performed in the project.

STREAMLIT DASHBOARD SECTION

A **Streamlit dashboard** was developed to present the customer churn analysis in an interactive and user-friendly format. The dashboard combines the major outputs from the Python analysis, SQL analysis, and machine learning stages.

14.1 DASHBOARD OVERVIEW

The dashboard provides an interactive view of customer churn and retention information. It allows users to examine important KPIs, customer segments, usage patterns, and predicted-risk information without directly working with the Python code or SQL queries.

The dashboard was developed using **Streamlit** and is contained in the app.py file.

14.2 KPI CARDS

The dashboard displays important customer-level KPIs, including:

- **Total Customers**
- **Churned Customers**
- **Churn Rate**
- **Retention Rate**

These KPIs provide a quick overview of the overall customer base and churn status.

14.3 CHURN ANALYSIS

The dashboard presents churn information across different customer characteristics. The available analysis includes:

- Churn by telecom partner
- Churn by age group
- Churn by gender

These visualizations make it easier to examine differences in observed churn patterns across customer segments.

14.4 PREDICTED-RISK SEGMENTS

The dashboard displays the three relative predicted-risk categories generated from the machine learning stage:

- Lower Predicted Risk
- Medium Predicted Risk
- Higher Predicted Risk

This allows users to view the distribution of customers across the predicted-risk segments.

14.5 USAGE ANALYSIS

Customer usage information is presented in the dashboard to support analysis of:

- Calls made
- SMS sent
- Data used

This connects customer activity information with the broader churn analysis.

14.6 HIGHER PREDICTED-RISK CUSTOMER TABLE

The dashboard also provides a customer-level table containing records from the **Higher Predicted Risk** segment.

This allows the user to examine individual customer information and use the available characteristics for further investigation.

14.7 INTERACTIVE FILTERS

Filters are provided for selected customer attributes, allowing users to narrow the displayed results and examine specific customer groups.

This makes the dashboard more flexible than a static report because users can interact with the available data.

14.8 RUNNING THE DASHBOARD

The Streamlit application can be launched from the project folder using:

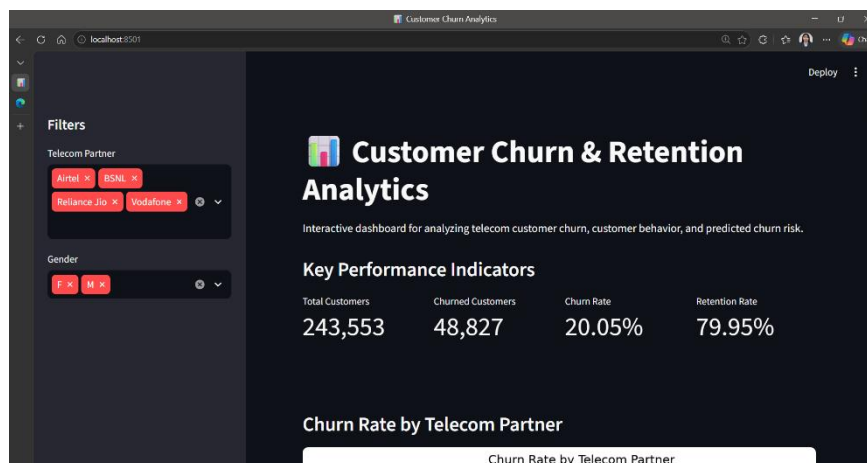
```
streamlit run app.py
```

After execution, Streamlit opens the dashboard in a web browser for interactive exploration.

14.9 PURPOSE OF THE DASHBOARD

The main purpose of the dashboard is to bring together the project's analytical outputs in a single interface. It connects **KPI analysis, churn analysis, machine learning predictions, risk segmentation, and customer-level information** into a format that is easier to explore and understand.

Figure: Streamlit Dashboard



RESULTS AND BUSINESS INSIGHTS

The Customer Churn & Retention Analytics project successfully analyzed **243,553 customer records across 14 variables** to identify churn patterns, customer behavior differences, and relative predicted-risk segments.

The major results and business insights obtained from the analysis are as follows:

1. **Data Quality and Preparation:**

The dataset was inspected for missing values, duplicate records, invalid values, and inconsistent data types. A negative value identified in the data_used field was treated as invalid and replaced using the median value. The cleaned dataset was then used for further analysis, machine learning, SQL analysis, and dashboard development.

2. **Churn Pattern Analysis:**

Customer churn was analyzed across multiple dimensions, including telecom partner, gender, age group, number of dependents, state, calls made, SMS sent, and data usage. This helped identify differences between churned and retained customer groups.

3. **Customer Behavior Insights:**

Numerical variables such as calls made, SMS sent, data usage, estimated salary, age, and number of dependents were compared between churned and retained customers. These comparisons provide useful descriptive information about customer behavior associated with churn.

4. **Telecom Partner and Geographic Analysis:**

Churn rates were examined across telecom partners and states. This allows organizations to identify areas or customer groups where churn monitoring may require additional attention.

5. **Machine Learning Results:**

Machine learning models were developed to estimate the probability that a customer may churn. Logistic Regression and Random Forest were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. The actual model evaluation results are stored in the model_comparison.csv file.

6. **Relative Customer Risk Segmentation:**

Customers were divided into **Lower Predicted Risk, Medium Predicted Risk, and Higher Predicted Risk** categories based on their predicted churn probabilities. The three groups contained **81,185, 81,184, and 81,184 customers**, respectively. These categories represent relative risk within the analyzed customer population and should not be interpreted as guaranteed churn outcomes.

7. **SQL-Based Business Analysis:**

SQL queries were used to calculate overall KPIs, churn by telecom partner and gender, state-level churn, usage differences, and predicted-risk segments. This provided a structured way to retrieve and summarize business information from the SQLite database.

8. Interactive Dashboard:

The Streamlit dashboard combines KPIs, churn analysis, customer behavior information, predicted-risk segments, and customer-level risk information in an interactive interface. Filters allow users to explore the results by selected customer characteristics.

Business Insights

The analysis provides several practical insights for retention planning:

- **Customer segmentation:** Customers can be grouped according to their observed characteristics and predicted churn risk.
- **Targeted retention:** Customers in the Higher Predicted Risk segment can be considered for proactive retention activities.
- **Usage-based engagement:** Calls, SMS, and data-usage patterns can be monitored to identify changes in customer engagement.
- **Partner-level monitoring:** Churn KPIs can be compared across telecom partners to identify differences that may require further investigation.
- **Continuous KPI monitoring:** Churn rate, retention rate, customer counts, and risk-segment distribution can be monitored regularly through the dashboard.
- **Data-driven decision support:** Combining Python analysis, SQL queries, machine learning, and Streamlit provides a complete workflow from raw customer data to actionable retention information.

These findings should be treated as **evidence for further investigation rather than proof of causation**. The analysis identifies patterns and predicted risk, but additional factors such as customer complaints, service quality, pricing, contract details, and customer-support interactions would be useful for understanding the underlying reasons for churn.

BUSINESS RECOMMENDATIONS

Based on the churn analysis, machine learning predictions, SQL analysis, and customer risk segmentation, the following business recommendations can be considered for improving customer retention:

- 1. Targeted Retention for Higher Predicted Risk Customers**
Customers classified under the **Higher Predicted Risk** segment can be prioritized for proactive retention activities such as personalized offers, service follow-ups, or loyalty benefits.
- 2. Customer Segmentation**
Customers should be segmented using demographic, usage, telecom-partner, and predicted-risk information. Different customer groups can then be addressed with appropriate retention strategies instead of using a single approach for all customers.
- 3. Usage-Based Customer Engagement**
Calls made, SMS sent, and data usage should be monitored to understand customer engagement patterns. Changes or lower levels of usage can be considered as signals for further investigation and customer engagement initiatives.
- 4. Telecom Partner-Level Monitoring**
Churn rates should be regularly compared across telecom partners. Where substantial differences are observed, the underlying reasons should be investigated before implementing specific corrective actions.
- 5. Continuous KPI Monitoring**
Important indicators such as **total customers, churned customers, churn rate, retention rate, and predicted-risk distribution** should be monitored regularly through the Streamlit dashboard.
- 6. Improve Customer Data Collection**
Additional variables such as customer complaints, network/service quality, pricing, contract details, customer-support interactions, and satisfaction information could be collected. These variables may provide additional information for understanding and addressing customer churn.
- 7. Use Predictions as Decision Support**
Machine learning predictions should be used to support retention planning rather than as guaranteed predictions of individual customer behavior. Retention teams can combine predicted risk with customer context before taking action.

Recommended Business Action Flow

Identify Higher Predicted Risk Customers → Analyze Customer Profile and Usage → Select Appropriate Retention Action → Monitor Response → Update Churn KPIs

This approach connects the project's **Data → Information → Insights → Decision → Action** framework with practical customer-retention activities.

LIMITATIONS

Although the project provides an end-to-end framework for customer churn analysis and retention planning, several limitations should be considered:

- 1. Limited Dataset Variables:**
The dataset contains customer demographic, usage, salary, geographic, and telecom-partner information, but it does not include several potentially important churn factors.
- 2. Missing Potential Churn Drivers:**
Variables such as customer complaints, network/service quality, pricing, contract details, customer-support interactions, and customer satisfaction are not available in the dataset. Their absence may limit the ability to explain the underlying reasons for churn.
- 3. No Causal Interpretation:**
The analysis identifies relationships and patterns between customer characteristics and churn. However, these relationships should not be interpreted as proof that a particular factor directly causes customer churn.
- 4. Machine Learning Prediction Limitations:**
The machine learning model produces estimated churn probabilities based on the available features. These predictions are not guarantees that an individual customer will churn.
- 5. Relative Risk Segmentation:**
The Lower, Medium, and Higher Predicted Risk categories are created relative to the analyzed customer population. Therefore, they represent comparative predicted risk rather than fixed or absolute definitions of customer churn risk.
- 6. Model Performance Depends on Available Data:**
The usefulness of the prediction model depends on the quality, relevance, and completeness of the available customer information. Additional business variables may improve future model performance.
- 7. Changing Customer Behavior:**
Customer behavior and engagement patterns can change over time. Therefore, the model and risk segments may need to be periodically updated using newer customer data.

These limitations should be considered when interpreting the analytical results and applying them to real-world customer-retention decisions.

FUTURE SCOPE

The Customer Churn & Retention Analytics project can be further enhanced in the following areas:

- 1. Advanced Model Optimization:**
Hyperparameter tuning and additional machine learning algorithms can be explored to improve churn prediction performance.
- 2. Feature Importance and Explainable AI:**
Techniques such as feature importance and SHAP analysis can be implemented to better understand which customer attributes contribute to model predictions.
- 3. Customer Lifetime Value (CLV):**
Customer Lifetime Value analysis can be integrated with churn probability to help prioritize retention efforts based on both churn risk and customer value.
- 4. Time-Based Churn Analysis:**
Future versions can analyze customer behavior over time to identify changes in engagement before churn occurs.
- 5. Additional Customer Information:**
Data related to customer complaints, service quality, pricing, contracts, support interactions, and satisfaction can be incorporated to improve churn analysis.
- 6. Automated Dashboard Updates:**
The Streamlit dashboard can be connected to regularly updated data so that KPIs, predictions, and risk segments are refreshed automatically.
- 7. Automated Retention Recommendations:**
A future system could generate personalized retention recommendations based on customer characteristics, predicted churn risk, usage behavior, and customer value.
- 8. Deployment and Scalability:**
The project can be deployed as a cloud-based analytics application so that authorized business users can access the dashboard and insights from different locations.

Overall, these enhancements can transform the current analytical project into a more comprehensive and continuously updated **customer retention decision-support system**.

CONCLUSION

The **Customer Churn & Retention Analytics** project successfully developed an end-to-end analytical solution for understanding customer churn and supporting customer retention planning. The project analyzed customer demographic, geographic, telecom-partner, and usage-related information to identify churn patterns and differences between churned and retained customers.

Python was used for **data cleaning, exploratory data analysis, visualization, KPI calculation, and machine learning**, while SQL was used to perform structured analysis of customer and risk-segment information. A machine learning model was also implemented to estimate customer churn probability and classify customers into **Lower Predicted Risk, Medium Predicted Risk, and Higher Predicted Risk** segments.

The Streamlit dashboard brings these analytical results together in an interactive interface, allowing users to examine key KPIs, churn patterns, customer usage information, and predicted-risk segments.

Overall, the project demonstrates the complete **Data → Information → Insights → Decision → Action** workflow. The resulting analysis can support data-driven customer-retention planning while providing a foundation for future improvements such as additional customer information, advanced machine learning techniques, explainable AI, customer lifetime value analysis, and automated dashboard updates.

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